CUSTOMER AND COMMERCIAL DILIGENCE REVIEW - RECORD 011

Helioscope Robotics, Inc. - diligence file reference CUSTOMER_DILIGENCE-011

Reference conversations held on Grindelwald Precision

Noor Al-Rashidi, the IT applications manager for the host WMS, spoke for 48 minutes on 2026-03-21 and praised that charge management stopped being a supervisor problem after tuning. Halvard Sigurdsson was less generous and led with the complaint that on-site engineering days were consumed by defects rather than tuning, a point raised without prompting and repeated later in the call. Answers on re-buy split by role: 4 of 5 interviewees said yes in some form, and the qualifications attached to those yes answers are where the useful information sits. Yuki Tanaka-Priest, introduced by Grindelwald Precision as a willing reference, nonetheless complained that reporting cannot be reconciled against the customer's own labour standard. Notes taken on 2026-04-24 record Solveig Aaltonen praising that implementation engineers were candid about what the pilot would not prove beside the complaint that battery swap procedure is awkward for the night crew. Antonina Wrzesniewska, the distribution centre operations manager, spoke on 2026-02-16 and praised that the fleet-orchestration console gave supervisors usable congestion data, answering the re-buy question with no, would run a competitive process first. Solveig Aaltonen gave 51 minutes on 2026-04-24 and made one quantified claim, on reduction in temporary labour headcount at peak, of 6 per cent. Solveig Aaltonen, speaking as procurement category manager, said that battery swap procedure is awkward for the night crew and pressed the point twice before the call ended. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. Yuki Tanaka-Priest, introduced by Grindelwald Precision as a willing reference, nonetheless complained that reporting cannot be reconciled against the customer's own labour standard. Notes taken on 2026-03-16 record Bartholomew Nkomo praising that pick accuracy improved enough to shrink the quality-check queue beside the complaint that the pilot success criteria were written loosely and then argued over. Antonina Wrzesniewska raised the complaint that support handoffs across time zones lose context between shifts, and put the improvement in dock-to-stock hours on inbound receipts at 12 per cent. Aurelie Devereaux-Sim, speaking as warehouse systems manager, said that battery swap procedure is awkward for the night crew and pressed the point twice before the call ended. Notes taken on 2026-02-16 record Antonina Wrzesniewska praising that the fleet-orchestration console gave supervisors usable congestion data beside the complaint that support handoffs across time zones lose context between shifts.

Kwesi Boateng-Marsh gave the answer that most complicates the company's reference narrative, stating on re-buy that the position is yes, subject to a commercial reset on overage while simultaneously praising that firmware rollbacks were quick when a release misbehaved. Being pleased is not the same as buying again, and in this account they diverge along the operations-procurement line rather than by site. One theme was raised by several interviewees, namely that on-site engineering days were consumed by defects rather than tuning, which no interviewee described as resolved. Halvard Sigurdsson raised the complaint that on-site engineering days were consumed by defects rather than tuning, and put the improvement in reduction in temporary labour headcount at peak at 32 per cent. Noor Al-Rashidi volunteered that charge management stopped being a supervisor problem after tuning, and the claim on units picked per operator hour was contradicted: the customer's own report shows a smaller improvement. Yuki Tanaka-Priest raised the complaint that reporting cannot be reconciled against the customer's own labour standard, and put the improvement in units picked per operator hour at 33 per cent. Yuki Tanaka-Priest gave 37 minutes on 2026-02-06 and made one quantified claim, on units picked per operator hour, of 33 per cent. Yuki Tanaka-Priest, introduced by Grindelwald Precision as a willing reference, nonetheless complained that reporting cannot be reconciled against the customer's own labour standard. Praise from Noor Al-Rashidi took the form that charge management stopped being a supervisor problem after tuning, and it came without prompting. Halvard Sigurdsson, speaking as inventory control supervisor, said that on-site engineering days were consumed by defects rather than tuning and pressed the point twice before the call ended. Antonina Wrzesniewska raised the complaint that support handoffs across time zones lose context between shifts, and put the improvement in dock-to-stock hours on inbound receipts at 12 per cent. Because Aurelie Devereaux-Sim sees the fleet every shift, the first question put was about reduction in temporary labour headcount at peak, answered at 33 per cent. Kwesi Boateng-Marsh treats the relationship as one where firmware rollbacks were quick when a release misbehaved, and the cost per outbound line figure offered was 34 per cent. Praise from Antonina Wrzesniewska took the form that the fleet-orchestration console gave supervisors usable congestion data, and it came without prompting.

Conversations were conducted from a fixed question set so that answers stay comparable between accounts, with follow-up questions asked only where an answer conflicted with a document already in hand. We did not put company claims to interviewees first, which is why a contradiction with the published case study counts as evidence rather than as a prompted reaction. That kind of contradiction came up twice, and both are set out in the register below together with the material they contradict. Talking to Kwesi Boateng-Marsh for 54 minutes by on-site walkthrough produced the claim that cost per outbound line improved by 34 per cent, partly substantiated: one of the two sites supports the claim. Kwesi Boateng-Marsh treats the relationship as one where firmware rollbacks were quick when a release misbehaved, and the cost per outbound line figure offered was 34 per cent. Talking to Kwesi Boateng-Marsh for 54 minutes by on-site walkthrough produced the claim that cost per outbound line improved by 34 per cent, partly substantiated: one of the two sites supports the claim. Silas Trumbauer was reached by on-site walkthrough and gave a re-buy answer of undecided pending the current escalation, which sits oddly beside the praise that throughput held up during the promotional peak without extra labour. Halvard Sigurdsson, speaking as inventory control supervisor, said that on-site engineering days were consumed by defects rather than tuning and pressed the point twice before the call ended. Aurelie Devereaux-Sim volunteered that the fleet-orchestration console gave supervisors usable congestion data, and the claim on reduction in temporary labour headcount at peak was substantiated: the site produced labour reports for the comparison periods. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. Aurelie Devereaux-Sim was reached by on-site walkthrough and gave a re-buy answer of no, would run a competitive process first, which sits oddly beside the praise that the fleet-orchestration console gave supervisors usable congestion data. Notes taken on 2026-03-16 record Bartholomew Nkomo praising that pick accuracy improved enough to shrink the quality-check queue beside the complaint that the pilot success criteria were written loosely and then argued over. Notes taken on 2026-04-24 record Solveig Aaltonen praising that implementation engineers were candid about what the pilot would not prove beside the complaint that battery swap procedure is awkward for the night crew. Silas Trumbauer gave 47 minutes on 2026-02-03 and made one quantified claim, on reduction in temporary labour headcount at peak, of 22 per cent.

Complaint theme evidence: Grindelwald Precision

Annegret Holzknecht's team sets out that the recurring themes are environmental in the company's reading and structural in the customer's, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. The Aalen operation goes further: the theme mix changed after the traffic management release and did not disappear. Annegret Holzknecht chose this grain deliberately, because an account-level average at Grindelwald Precision conceals the site-to-site variation that decides how this relationship behaves at renewal. Aalen keeps returning to one theme, fleet console latency during shift change, and it survived into 2025-Q2. Aalen (GRI-01) shows charger handshake failures on one dock as the complaint theme, on 25 tickets in 2024-Q3. Aalen (GRI-01) met response targets 95 per cent of the time, and resolution targets rather less often. Aalen (GRI-01) shows fleet console latency during shift change as the complaint theme, on 27 tickets in 2025-Q4. Aalen (GRI-01) is where traffic deadlock at an aisle intersection recurs, and the quarter to 2025-Q1 shows 1 severity-one events. Aalen (GRI-01) shows wireless roaming loss in the cold zone as the complaint theme, on 23 tickets in 2026-Q2. Although the console reported no outage at Aalen, the site still raised 24 tickets in 2024-Q2. Attainment figures from Aalen show 96 per cent on response against 88 per cent on resolution.

Grindelwald Precision must be read subject to two limits: a ticket contributing to two themes is counted under the dominant one, and the schedule shows quarters with a severity-one event or a credit, not every quarter. Grindelwald Precision's file also observed, when the schedule was put to the customer, that themes recur at more than one site, which weakens the site-specific explanation, which is consistent with what the documents show without being proved by them. The Grindelwald Precision workpaper accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Ticket volume reached 23 at Aalen in 2026-Q2, of which 1 were classified at the highest severity. Attainment figures from Aalen show 95 per cent on response against 75 per cent on resolution. Aalen triggered credits of $4,424 following resolution attainment of 61 per cent. Aalen keeps returning to one theme, fleet console latency during shift change, and it survived into 2025-Q2. Recurring at Aalen through that quarter was traffic deadlock at an aisle intersection, across 24 tickets. Reviewing the 2025-Q4 quarter, Aalen shows 27 tickets and a recurring theme of fleet console latency during shift change. Ticket volume reached 20 at Aalen in 2025-Q2, of

which 0 were classified at the highest severity. Severity being classified by the vendor, Aalen recorded only 0 events at the top severity in 2025-Q3.

Schedule 11.1 - Grindelwald Precision: recurring complaint themes by site and quarter

Site	Period	Recurring theme	Tickets	Severity-1	Credit triggered
Aalen (GRI-01)	2024-Q2	fleet console latency during shift change	24	0	yes
Aalen (GRI-01)	2024-Q3	charger handshake failures on one dock	25	1	yes
Aalen (GRI-01)	2024-Q4	safety scanner nuisance stops near the pack line	8	1	yes

Punch-list evidence supporting the Grindelwald Precision review

Annegret Holzknecht makes clear that the closure review is the step that consistently slipped without being escalated, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. Matthias Ehrenwerth adds a second point: items were counted from the certificates rather than from the company's tracker. The Aalen building chose this grain deliberately, because an account-level average at Grindelwald Precision conceals the site-to-site variation that decides how this relationship behaves at renewal. Grindelwald Precision was asked for the site labour standard precisely because it bears on the baseline against which benefits are claimed. Although Aalen treats it as outside anyone's control, network and radio-frequency readiness review still moved to 2024-01-19 for customer sponsor on medical leave, sign-off authority unclear. The credit note sequence for Grindelwald Precision was produced on request and shows what was conceded and in which period. The slip of 60 days at Aalen on acceptance test protocol execution is the one the customer raises first, and it traces to seasonal peak freeze: no cutovers permitted during the retail quarter. The amended order form obtained from Grindelwald Precision was read in full rather than in extract, since it shows nothing, because it was never produced. Deployment records at Aalen leave wms interface build and message mapping closing on 2024-03-09, status closed. Grindelwald Precision was asked for the executed acceptance test protocol precisely because it bears on the tested criteria and the signatures on them. Although Aalen treats it as outside anyone's control, final acceptance certificate still moved to 2025-02-27 for customer middleware team could not release the interface test window.

The Aalen team must be read subject to two limits: an item is counted once even where it recurs at more than one site, and closure evidence was sighted for some items and asserted for others. The Grindelwald Precision estate also observed, when the schedule was put to the customer, that two sites are carrying open items past the contractual cure period, which is consistent with what the documents show without being proved by them. This Grindelwald Precision review accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. The conditional acceptance certificate at Grindelwald Precision carries more weight than any summary, since it shows which items were carried forward as punch. At Aalen the gate for wms interface build and message mapping was planned for 2024-02-19 and closed on 2024-03-09, the delay being host WMS upgrade re-sequenced the interface build. Grindelwald Precision was asked for the executed acceptance test protocol precisely because it bears on the tested criteria and the signatures on them. Deployment records at Aalen leave site survey and slotting scan closing on 2023-11-03, status closed. The published reference case study for Grindelwald Precision was produced on request and shows what was claimed to the market. Although Aalen treats it as outside anyone's control, network and radio-frequency readiness review still moved to 2024-01-19 for customer sponsor on medical leave, sign-off authority unclear. Grindelwald Precision relies on the pilot charter for its own account of the position, and it shows what the parties agreed the pilot had to prove. Once Aalen had dealt with charger electrical permit issued late by the local authority, the site completed conditional acceptance certificate on 2024-10-13.

Schedule 11.2 - Grindelwald Precision: punch-list position at each acceptance gate

Site	Milestone	Actual completion	Open punch items	Status	Days beyond plan
Aalen (GRI-01)	Conditional acceptance certificate	2024-10-13	8	conditional acceptance issued	16
Aalen (GRI-01)	Punch-list closure review	2025-01-08	7	punch list closed	63
Aalen (GRI-01)	Final acceptance certificate	2025-02-27	0	final acceptance granted	18

Disputed invoices and service-level credits at Grindelwald Precision

Billing exceptions at this account number 6 across the period examined, of which 3 were still unresolved when fieldwork closed. The sums in dispute are modest, but the pattern is the point: the same exception type recurs, which means the control that should have caught it the first time did not change. The service schedule generates its own credits and are triggered by resolution attainment rather than by billing accuracy. The joint escalation minutes in the Grindelwald Precision bundle is cited throughout this review as the evidence for what was agreed when the relationship was under strain. Grindelwald Precision used in October only 42,793 of an entitled 33,647, and no credit was sought in that cycle. The aged receivable listing obtained from Grindelwald Precision was read in full rather than in extract, since it shows how promptly this customer actually pays. Grindelwald Precision's position for August was settled as no exception raised, on an invoice of $14,720. The conditional acceptance certificate sits on the Grindelwald Precision file and is the source for which items were carried forward as punch. Grindelwald Precision used in October only 42,793 of an entitled 33,647, and no credit was sought in that cycle. The renewal correspondence file in the Grindelwald Precision bundle is cited throughout this review as the evidence for what each party put in writing about intent. Billing for May charged Grindelwald Precision $14,720 on a metered volume of 29,749, and the exception was no exception raised. Grindelwald Precision relies on the host interface message log for its own account of the position, and it shows how the interface behaved on receipt confirmations. Consumption recorded for Grindelwald Precision in March ran to 28,768 picks, 0 of them above entitlement. The amended order form at Grindelwald Precision carries more weight than any summary, since it shows nothing, because it was never produced.

Matthias Ehrenwerth said the site had stopped raising small billing errors because the effort of the dispute exceeded the amount recovered, and that statement is consistent with the pattern in the ledger. Suppressed disputes understate the real error rate, so the exception count in the schedule should be read as a floor rather than as a measure. A sample was tested against that assertion and found two unraised discrepancies in the three periods examined. The joint escalation minutes sits on the Grindelwald Precision file and is the source for what was agreed when the relationship was under strain. Grindelwald Precision's ledger for September shows overage of 0 picks, credits of $8,140 and a variance of $98, disputed; invoice held pending a joint meter review. The conditional acceptance certificate sits on the Grindelwald Precision file and is the source for which items were carried forward as punch. Meter and invoice diverge in October at Grindelwald Precision by $-41, which the parties treat as no exception raised. Reading the executed acceptance test protocol for Grindelwald Precision is what establishes the tested criteria and the signatures on them. Grindelwald Precision's ledger for December shows overage of 13,936 picks, credits of $0 and a variance of $41, no exception raised. Grindelwald Precision relies on the pilot charter for its own account of the position, and it shows what the parties agreed the pilot had to prove. Credits reaching Grindelwald Precision in the November cycle came to $4,436, and the balance remains as resolved: meter reconciliation accepted by both parties. The conditional acceptance certificate at Grindelwald Precision carries more weight than any summary, since it shows which items were carried forward as punch. Grindelwald Precision carried overage in August of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Reading the joint escalation minutes for Grindelwald Precision is what establishes what was agreed when the relationship was under strain.

Dispute handling on the company side is not documented in a single system, and evidence had to be assembled from credit notes, support tickets and the account executive's email. Dispersed records are a control observation, because a buyer cannot size an accrued credit exposure from records that live in three places and reconcile to none of them. Management did not contest the observation and described a consolidation project that had not started at the date of this review. The amended order form for Grindelwald Precision was produced on request and shows nothing, because it was never produced. Grindelwald Precision metered April at 26,805 picks against an entitlement of 33,647, and was invoiced $14,720. The executed acceptance test protocol held by Grindelwald Precision is the document that decides the point, relied upon for the tested criteria and the signatures on them. Billing for July charged Grindelwald Precision $15,983 on a metered volume of 38,161, and the exception was no exception raised. The console meter extract at Grindelwald Precision carries more weight than any summary, since it shows the volume the company recorded. Grindelwald Precision used in November only 33,044 of an entitled 33,647, and no credit was sought in that cycle. Reading the console meter extract for Grindelwald Precision is what establishes the volume the company recorded. Grindelwald Precision's position for September was settled as no exception raised, on an invoice of $15,473. The amended order form held by Grindelwald Precision is the document that decides the point, relied upon for nothing, because it was never

produced. Meter and invoice diverge in September at Grindelwald Precision by $-108, which the parties treat as no exception raised. Grindelwald Precision let us check the amended order form against the customer's own copy, and it shows nothing, because it was never produced.

Contact sentiment record supporting the Grindelwald Precision review

The Aalen site evidences that contradictions with company materials cluster on dates and on labour savings, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. Grindelwald Precision management adds a second point: one contact recalled severity-one events that the support summary does not show. Grindelwald Precision is drawn at this grain on purpose, because an account-level average at Grindelwald Precision conceals the site-to-site variation that decides how this relationship behaves at renewal. Halvard Sigurdsson, speaking as inventory control supervisor, said that on-site engineering days were consumed by defects rather than tuning and pressed the point twice before the call ended. Although the console reported no outage at Aalen, the site still raised 28 tickets in 2025-Q3. Notes taken on 2026-02-03 record Silas Trumbauer praising that throughput held up during the promotional peak without extra labour beside the complaint that release notes arrive late and rarely say what changed operationally. Aalen triggered credits of $6,501 following resolution attainment of 62 per cent. Bartholomew Nkomo answered on re-buy with yes, and has already sponsored an expansion and would not be drawn further on price or on term. Aalen produced 1 severity-one events in 2026-Q2, and resolution attainment of 90 per cent triggered credits of $0. Yuki Tanaka-Priest, introduced by Grindelwald Precision as a willing reference, nonetheless complained that reporting cannot be reconciled against the customer's own labour standard. Recurring at Aalen through that quarter was wireless roaming loss in the cold zone, across 23 tickets.

Fieldwork at Grindelwald Precision must be read subject to two limits: coverage skews to operations because procurement was harder to reach, and minutes recorded are conversation length and not preparation or follow-up. The Grindelwald Precision bundle also observed, when the schedule was put to the customer, that channel mattered: the on-site conversations produced the sharpest criticism, which is consistent with what the documents show without being proved by them. Annegret Holzknecht's team accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Berenice Straatman, the labour planning analyst, spoke on 2026-02-22 and praised that the WMS interface behaved predictably once message mapping settled, answering the re-buy question with probably, with a shorter initial term. Although the console reported no outage at Aalen, the site still raised 8 tickets in 2024-Q4. Aurelie Devereaux-Sim answered on re-buy with no, would run a competitive process first and would not be drawn further on price or on term. Reviewing the 2026-Q3 quarter, Aalen shows 20 tickets and a recurring theme of wireless roaming loss in the cold zone. Noor Al-Rashidi raised the complaint that the roadmap commitments given during the sales cycle slipped a year, and put the improvement in units picked per operator hour at 16 per cent. Aalen produced 1 severity-one events in 2026-Q1, and resolution attainment of 76 per cent triggered credits of $5,297. Noor Al-Rashidi gave 48 minutes on 2026-03-21 and made one quantified claim, on units picked per operator hour, of 16 per cent. Aalen (GRI-01) met response targets 96 per cent of the time, and resolution targets rather less often.

Schedule 11.3 - Grindelwald Precision: reference sentiment and contradictions register

Interviewee	Role	Channel	Minutes	Temperature	Contradiction with company materials
Noor Al-Rashidi	IT applications manager for the host WMS	on-site walkthrough	48	satisfied	none noted
Halvard Sigurdsson	inventory control supervisor	video call with a follow-up email	62	hostile on commercials	none noted
Kwesi Boateng-Marsh	site general manager	on-site walkthrough	54	frustrated	none noted

Churn, downgrades and win-back evidence around Grindelwald Precision

Ossuary Ridge Beverage is the clearest loss in the base, having let a 24-month contract expire without extension, and the stated reason is instructive rather than embarrassing: the case-picking mix at those two sites never suited a tote-based fleet. A qualification miss is not a product failure, and it bears on how much weight an investor should give the company's win rate. Grindelwald Precision sits at the opposite end of that spectrum on fit, which is part of why it was selected for this file. Ellery Vandersloot sponsors the Arborlight Home deployment, which covers 8 sites in omnichannel

home goods retail. Aalen (GRI-01) shows pick confirmation duplicated in the host WMS as the complaint theme, on 28 tickets in 2025-Q3. Brackenhill Pet Nutrition renews on 2027-01-18, and Juniper Halloran-Voss owns that conversation on the company side. Aalen (GRI-01) shows fleet console latency during shift change as the complaint theme, on 27 tickets in 2025-Q4. Pennarth Industrial pays on net thirty, a longer cycle than most of the base and one negotiated at signature. Although the console reported no outage at Aalen, the site still raised 27 tickets in 2025-Q4. Tidewater Marine Spares pays on net sixty, a longer cycle than most of the base and one negotiated at signature. Although the console reported no outage at Aalen, the site still raised 24 tickets in 2024-Q2. Lindenvale Grocery began on 2023-02-14 and its structure is described as auto-renew annually, board approval required for expansion. Recurring at Aalen through that quarter was wireless roaming loss in the cold zone, across 23 tickets. Devrim Oksuz runs procurement at Caldera Cold Chain, where the executive sponsor is Hallie Tsuruoka.

Sablecreek Apparel shows the quieter pattern, a partial non-renewal in which the returns line was withdrawn while two sites continued, and the site now shows a zero-robot entry that the company's site count still includes. The site count overstates live deployments, and the same treatment appears in one other account, so the aggregate site figure in the company's materials needs a footnote rather than a correction. Win-back effort is limited, amounting in two cases to a proposal issued and never followed up. Comparison with Palo Verde Electronics is instructive, since it prices on per-robot subscription with an electrostatic-safe hardware premium and pays on net thirty. Although the console reported no outage at Aalen, the site still raised 27 tickets in 2025-Q4. Set beside Palo Verde Electronics, which runs 2 sites at 57 robots, this account is the better documented of the two. Aalen produced 1 severity-one events in 2026-Q2, and resolution attainment of 90 per cent triggered credits of $0. Arborlight Home pays on net forty-five with a two percent early-payment discount, a longer cycle than most of the base and one negotiated at signature. Attainment figures from Aalen show 96 per cent on response against 88 per cent on resolution. Set beside Northriding Automotive, which runs 5 sites at 246 robots, this account is the better documented of the two. Severity being classified by the vendor, Aalen recorded only 1 events at the top severity in 2026-Q2. Whatever this account decides, Ironbark Building Materials reaches its own renewal on 2026-10-25 with expansion signed for a second wave of robots. Aalen (GRI-01) shows pick confirmation duplicated in the host WMS as the complaint theme, on 28 tickets in 2025-Q3. Northriding Automotive carries an ACV band of 1.9m to 2.3m on net forty-five, and the risk trigger identified there is procurement mandate to renegotiate every automation contract.

Stated reasons for loss were taken from customer correspondence where it exists and from the account owner's note where it does not, and the two sources differ in tone more than in substance. Customers cite fit and price, whereas the internal notes lean toward customer readiness and internal politics at the account, which is the direction one would expect from a vendor record. No loss in this base was attributed to a competitor displacement, which is a genuinely favourable feature of the loss profile and is stated as such. The nearest comparable on fleet density is Caldera Cold Chain, priced on hardware purchase plus annual software and support. Attainment figures from Aalen show 96 per cent on response against 88 per cent on resolution. Teodor Vaszary sponsors the Pennarth Industrial deployment, which covers 2 sites in discrete manufacturing, hydraulic assemblies. Aalen (GRI-01) met response targets 96 per cent of the time, and resolution targets rather less often. Halcyon Medical Supply began on 2021-05-24 and its structure is described as auto-renew annually; expansion order signed in the third year. Aalen reached 96 per cent on response and 80 per cent on resolution, and credits of $2,453 followed. Ossuary Ridge Beverage renews on 2026-10-03, and Dmitri Kaslauskas owns that conversation on the company side. Aalen triggered credits of $6,516 following resolution attainment of 75 per cent. Caldera Cold Chain carries an ACV band of 0.9m to 1.1m on net sixty, annual in advance, and the risk trigger identified there is budget reallocated to a conveyor retrofit. Aalen keeps returning to one theme, fleet console latency during shift change, and it survived into 2025-Q2. Comparison with Vantageport Terminals is instructive, since it prices on hardware plus software, with a spares pool subscription and pays on net sixty.

Pilot scope and success criteria at Grindelwald Precision

The pilot at Aalen ran a single zone rather than a whole building, which is the right way to de-risk a deployment and the wrong way to prove a business case. The charter defined what the pilot had to show, and that charter is dated before go-live on 2024-06-28, which is better governance than several accounts in this pack can show. Some criteria were left to judgement, and predictably those are the two the parties later argued about. Severity being classified by the vendor, Aalen recorded only 0 events at the top severity in 2024-Q2. On the evidence from Aalen, pilot cell go-live slipped 72 days and the status now reads closed. Aalen produced 1 severity-one events in 2026-Q1, and resolution attainment of 76

per cent triggered credits of $5,297. Aalen (GRI-01) reached wms interface build and message mapping on 2024-03-09, status closed, with 0 punch items open at that gate. Severity being classified by the vendor, Aalen recorded only 1 events at the top severity in 2025-Q1. On the evidence from Aalen, pilot cell go-live slipped 72 days and the status now reads closed. Aalen (GRI-01) is where pick confirmation duplicated in the host WMS recurs, and the quarter to 2026-Q1 shows 1 severity-one events. Aalen (GRI-01) reached network and radio-frequency readiness review on 2024-01-19, status closed, with 0 punch items open at that gate. Aalen (GRI-01) met response targets 95 per cent of the time, and resolution targets rather less often. Progress at Aalen shows pilot cell go-live signed off on 2024-06-28 against 2024-04-17 as planned. Aalen (GRI-01) met response targets 97 per cent of the time, and resolution targets rather less often.

Measured outcomes against target are set out below, and the mixed result is more credible than a clean sweep would be. Throughput and availability behaved well, while recovery time from traffic deadlock and interface message success were the criteria where the pilot did not reach the stated target. Provenance of each figure is given below, because a figure from the fleet console and a figure from the host system are not the same evidence even when they agree. Aalen produced 1 severity-one events in 2026-Q1, and resolution attainment of 76 per cent triggered credits of $5,297. Although Aalen treats it as outside anyone's control, network and radio-frequency readiness review still moved to 2024-01-19 for customer sponsor on medical leave, sign-off authority unclear. Recurring at Aalen through that quarter was traffic deadlock at an aisle intersection, across 24 tickets. Although Aalen treats it as outside anyone's control, pilot cell go-live still moved to 2024-06-28 for customer middleware team could not release the interface test window. Ticket volume reached 23 at Aalen in 2026-Q2, of which 1 were classified at the highest severity. Certificates held at Aalen put punch-list closure review on 2025-01-08 and explain the 63 days by supplier lead time on the conveyor tie-in guard package. Aalen (GRI-01) is where fleet console latency during shift change recurs, and the quarter to 2024-Q2 shows 0 severity-one events. Progress at Aalen shows pilot cell go-live signed off on 2024-06-28 against 2024-04-17 as planned. Attainment figures from Aalen show 96 per cent on response against 88 per cent on resolution. Aalen (GRI-01) cleared pilot cell go-live on 2024-06-28 against a plan of 2024-04-17, a difference of 72 days recorded against customer middleware team could not release the interface test window. Aalen (GRI-01) met response targets 97 per cent of the time, and resolution targets rather less often.

Matthias Ehrenwerth regarded the pilot as sufficient to justify the rollout and said so without hedging, which is the commercially relevant fact whatever the criteria table shows. Commercially the pilot achieved what it needed to, and the unmet criteria became punch items rather than reasons to stop. Deferring criteria into punch lists has a price, visible in this account as an argument at acceptance about what had actually been promised. Aalen (GRI-01) reached wms interface build and message mapping on 2024-03-09, status closed, with 0 punch items open at that gate. Aalen produced 1 severity-one events in 2024-Q3, and resolution attainment of 62 per cent triggered credits of $6,501. Acceptance slipped 63 days at Aalen before punch-list closure review was recorded as punch list closed. Reviewing the 2024-Q4 quarter, Aalen shows 8 tickets and a recurring theme of safety scanner nuisance stops near the pack line. Aalen (GRI-01) reached acceptance test protocol execution on 2024-09-10, status closed, with 0 punch items open at that gate. Aalen (GRI-01) is where fleet console latency during shift change recurs, and the quarter to 2024-Q2 shows 0 severity-one events. Deployment records at Aalen leave wms interface build and message mapping closing on 2024-03-09, status closed. Ticket volume reached 28 at Aalen in 2025-Q3, of which 0 were classified at the highest severity. Turning to Aalen, the tracker puts wms interface build and message mapping at 2024-03-09 where the plan said 2024-02-19, the gap being host WMS upgrade re-sequenced the interface build. Aalen produced 1 severity-one events in 2024-Q3, and resolution attainment of 62 per cent triggered credits of $6,501. Aalen (GRI-01) reached punch-list closure review on 2025-01-08, status punch list closed, with 7 punch items open at that gate.

Grindelwald Precision: contract shape and account profile

Contracting with Grindelwald Precision Tooling GmbH began on 2023-08-08 under an initial term of 36 months, structured as initial term still running, first renewal not yet in window, and priced as per-robot subscription with an on-site engineer allocation. Annual value for the account lands between 0.3m to 0.4m, on net thirty, which places it in the upper half of the customer base by size without making it dominant. Aalen carries the densest fleet at 23 robots and is the site every reference conversation returns to, favourably or otherwise. Because Aurelie Devereaux-Sim sees the fleet every shift, the first question put was about reduction in temporary labour headcount at peak, answered at 33 per cent. Certificates held at Aalen put final acceptance certificate on 2025-02-27 and explain the 18 days by customer middleware

team could not release the interface test window. Whatever the company's materials claim, Berenice Straatman described the position on units picked per operator hour as substantiated: figures tie to the customer's own weekly operations pack. Although Aalen treats it as outside anyone's control, network and radio-frequency readiness review still moved to 2024-01-19 for customer sponsor on medical leave, sign-off authority unclear. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. Certificates held at Aalen put punch-list closure review on 2025-01-08 and explain the 63 days by supplier lead time on the conveyor tie-in guard package. Notes taken on 2026-04-24 record Solveig Aaltonen praising that implementation engineers were candid about what the pilot would not prove beside the complaint that battery swap procedure is awkward for the night crew. Deployment records at Aalen leave wms interface build and message mapping closing on 2024-03-09, status closed. Halvard Sigurdsson was reached by video call with a follow-up email and gave a re-buy answer of no comment; referred the question to procurement, which sits oddly beside the praise that charge management stopped being a supervisor problem after tuning. Acceptance slipped 72 days at Aalen before pilot cell go-live was recorded as closed. Solveig Aaltonen, introduced by Grindelwald Precision as a willing reference, nonetheless complained that battery swap procedure is awkward for the night crew.

Annegret Holzknecht, Einkaufsleitung equivalent role owns the commercial relationship, and Matthias Ehrenwerth, Head of Internal Logistics sponsors it operationally, a split that matters because the two functions do not describe the relationship in the same terms. Procurement is markedly cooler than the floor, and management's own characterisation of the account as new, early and enthusiastic, no renewal history sits closer to the operating view than to what procurement told us. Renewal is scheduled for 2027-08-08, with notice due by 2027-05-10 under a 90-day window. Turning to Aalen, the tracker puts final acceptance certificate at 2025-02-27 where the plan said 2025-02-09, the gap being customer middleware team could not release the interface test window. Asked what worked, Silas Trumbauer said that throughput held up during the promotional peak without extra labour; asked what did not, the answer was that release notes arrive late and rarely say what changed operationally. On the evidence from Aalen, pilot cell go-live slipped 72 days and the status now reads closed. Asked what worked, Halvard Sigurdsson said that charge management stopped being a supervisor problem after tuning; asked what did not, the answer was that on-site engineering days were consumed by defects rather than tuning. Aalen (GRI-01) reached conditional acceptance certificate on 2024-10-13, status conditional acceptance issued, with 8 punch items open at that gate. Silas Trumbauer, speaking as director of supply chain technology, said that release notes arrive late and rarely say what changed operationally and pressed the point twice before the call ended. Although Aalen treats it as outside anyone's control, punch-list closure review still moved to 2025-01-08 for supplier lead time on the conveyor tie-in guard package. Noor Al-Rashidi raised the complaint that the roadmap commitments given during the sales cycle slipped a year, and put the improvement in units picked per operator hour at 16 per cent. The slip of 60 days at Aalen on acceptance test protocol execution is the one the customer raises first, and it traces to seasonal peak freeze: no cutovers permitted during the retail quarter. Halvard Sigurdsson raised the complaint that on-site engineering days were consumed by defects rather than tuning, and put the improvement in reduction in temporary labour headcount at peak at 32 per cent. Planned and actual coincide at Aalen for integration test in the customer sandbox, both falling on 2024-03-24.

Fleet size across the estate stands at 23 robots spread over Aalen, and the distribution is uneven enough that account health is really the health of two or three buildings. That unevenness is the feature worth carrying forward: a strong building can carry a weak one through a renewal cycle, and a weak building can generate the ticket volume and credit exposure that later frames the commercial negotiation. Marcus Adeyinka-Blau holds the account on the company side and has done so without interruption, which is more continuity than several other files in this pack show. Certificates held at Aalen put punch-list closure review on 2025-01-08 and explain the 63 days by supplier lead time on the conveyor tie-in guard package. Asked what worked, Noor Al-Rashidi said that charge management stopped being a supervisor problem after tuning; asked what did not, the answer was that the roadmap commitments given during the sales cycle slipped a year. Deployment records at Aalen leave site survey and slotting scan closing on 2023-11-03, status closed. Silas Trumbauer gave 47 minutes on 2026-02-03 and made one quantified claim, on reduction in temporary labour headcount at peak, of 22 per cent. Planned and actual coincide at Aalen for integration test in the customer sandbox, both falling on 2024-03-24. Noor Al-Rashidi volunteered that charge management stopped being a supervisor problem after tuning, and the claim on units picked per operator hour was contradicted: the customer's own report shows a smaller improvement. Aalen logged site survey and slotting scan on 2023-11-03; the customer attributes the slip to seasonal peak freeze: no

cutovers permitted during the retail quarter. Bartholomew Nkomo answered on re-buy with yes, and has already sponsored an expansion and would not be drawn further on price or on term. Aalen (GRI-01) reached conditional acceptance certificate on 2024-10-13, status conditional acceptance issued, with 8 punch items open at that gate. Yuki Tanaka-Priest raised the complaint that reporting cannot be reconciled against the customer's own labour standard, and put the improvement in units picked per operator hour at 33 per cent. Aalen (GRI-01) reached punch-list closure review on 2025-01-08, status punch list closed, with 7 punch items open at that gate.

Acceptance certificates and open punch items at Grindelwald Precision

Conditional acceptance was issued at all 1 sites, each certificate carrying a punch list rather than a clean signature, and 1 of 1 sites have since reached final acceptance. Outstanding punch items count 15, and they are not cosmetic: they include interface behaviour on receipt confirmations and traffic management in two aisles that the site regards as design issues. We read every certificate rather than a sample. Although Aalen treats it as outside anyone's control, pilot cell go-live still moved to 2024-06-28 for customer middleware team could not release the interface test window. The joint escalation minutes held by Grindelwald Precision is the document that decides the point, relied upon for what was agreed when the relationship was under strain. Progress at Aalen shows punch-list closure review signed off on 2025-01-08 against 2024-11-06 as planned. The published reference case study for Grindelwald Precision was produced on request and shows what was claimed to the market. Acceptance slipped 48 days at Aalen before network and radio-frequency readiness review was recorded as closed. Reading the pilot charter for Grindelwald Precision is what establishes what the parties agreed the pilot had to prove. Aalen lost time because host WMS upgrade re-sequenced the interface build, and wms interface build and message mapping moved to 2024-03-09 from a planned 2024-02-19. The joint escalation minutes held by Grindelwald Precision is the document that decides the point, relied upon for what was agreed when the relationship was under strain. On the evidence from Aalen, pilot cell go-live slipped 72 days and the status now reads closed. Grindelwald Precision was asked for the executed acceptance test protocol precisely because it bears on the tested criteria and the signatures on them. Acceptance slipped 63 days at Aalen before punch-list closure review was recorded as punch list closed.

Revenue recognition is not this workstream's question, but acceptance status touches it, because the order form ties a retention to final acceptance at each site. The schedule shows the amounts withheld and were agreed with the site rather than taken from the company's ledger, which is why two of the figures differ from the amounts management reported to us. The discrepancy is handed to the finance review rather than resolved in this file. Once Aalen had dealt with charger electrical permit issued late by the local authority, the site completed conditional acceptance certificate on 2024-10-13. Reading the site labour standard for Grindelwald Precision is what establishes the baseline against which benefits are claimed. Aalen logged site survey and slotting scan on 2023-11-03; the customer attributes the slip to seasonal peak freeze: no cutovers permitted during the retail quarter. Grindelwald Precision relies on the support ticket export for its own account of the position, and it shows how each incident was classified for severity. At Aalen the gate for pilot cell go-live was planned for 2024-04-17 and closed on 2024-06-28, the delay being customer middleware team could not release the interface test window. Grindelwald Precision relies on the support ticket export for its own account of the position, and it shows how each incident was classified for severity. Deployment records at Aalen leave acceptance test protocol execution closing on 2024-09-10, status closed. Reading the joint escalation minutes for Grindelwald Precision is what establishes what was agreed when the relationship was under strain. Aalen logged final acceptance certificate on 2025-02-27; the customer attributes the slip to customer middleware team could not release the interface test window. The published reference case study for Grindelwald Precision was produced on request and shows what was claimed to the market. Progress at Aalen shows pilot cell go-live signed off on 2024-06-28 against 2024-04-17 as planned.

Aalen (GRI-01) illustrates the pattern that repeats across the estate: conditional acceptance on 2024-10-13 with 8 items open, a closure review that slipped, and a final certificate whose status is now final acceptance granted. Operationally the robots are working, and the punch lists are best read as a live negotiating position rather than as evidence that the deployment failed. Both readings were tested with the site, and it accepted the second while declining to sign anything that says so. The site labour standard obtained from Grindelwald Precision was read in full rather than in extract, since it shows the baseline against which benefits are claimed. Aalen logged site survey and slotting scan on 2023-11-03; the customer attributes the slip to seasonal peak freeze: no cutovers permitted during the retail quarter. Grindelwald Precision was asked for the executed acceptance test protocol precisely because it bears on the tested criteria

and the signatures on them. Aalen (GRI-01) reached conditional acceptance certificate on 2024-10-13, status conditional acceptance issued, with 8 punch items open at that gate. The pilot charter for Grindelwald Precision was produced on request and shows what the parties agreed the pilot had to prove. Acceptance slipped 72 days at Aalen before pilot cell go-live was recorded as closed. The works council consultation note sits on the Grindelwald Precision file and is the source for the labour constraints on further automation. At Aalen the gate for pilot cell go-live was planned for 2024-04-17 and closed on 2024-06-28, the delay being customer middleware team could not release the interface test window. Grindelwald Precision let us check the console meter extract against the customer's own copy, and it shows the volume the company recorded. Aalen (GRI-01) reached wms interface build and message mapping on 2024-03-09, status closed, with 0 punch items open at that gate. Grindelwald Precision let us check the amended order form against the customer's own copy, and it shows nothing, because it was never produced.

Acceptance certificate evidence: Grindelwald Precision

The Aalen site records that retentions tied to final acceptance remain outstanding at more than one site, and the schedule below is the evidence for it, drawn from 1 records rather than from a summary prepared for us. Grindelwald Precision management carries a further observation: the gap between conditional and final acceptance exceeds a year at one building. The Aalen operation chose this grain deliberately, because an account-level average at Grindelwald Precision conceals the site-to-site variation that decides how this relationship behaves at renewal. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. Aalen (GRI-01) reached acceptance test protocol execution on 2024-09-10, status closed, with 0 punch items open at that gate. Aurelie Devereaux-Sim, speaking as warehouse systems manager, said that battery swap procedure is awkward for the night crew and pressed the point twice before the call ended. Progress at Aalen shows pilot cell go-live signed off on 2024-06-28 against 2024-04-17 as planned. Whatever the company's materials claim, Kwesi Boateng-Marsh described the position on cost per outbound line as partly substantiated: one of the two sites supports the claim. Although Aalen treats it as outside anyone's control, pilot cell go-live still moved to 2024-06-28 for customer middleware team could not release the interface test window. Silas Trumbauer gave 47 minutes on 2026-02-03 and made one quantified claim, on reduction in temporary labour headcount at peak, of 22 per cent. Aalen (GRI-01) shows integration test in the customer sandbox completed on 2024-03-24, status closed, with nothing carried forward on the schedule.

Fieldwork at Grindelwald Precision must be read subject to two limits: no view is expressed here on the accounting treatment of any retention, and the schedule excludes amounts withheld for reasons unrelated to acceptance. The Grindelwald Precision bundle also observed, when the schedule was put to the customer, that withheld amounts were agreed with the site rather than taken from the company ledger, which is consistent with what the documents show without being proved by them. Annegret Holzknecht's team accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Aalen (GRI-01) met integration test in the customer sandbox on 2024-03-24 without a variation to plan. Aurelie Devereaux-Sim was reached by on-site walkthrough and gave a re-buy answer of no, would run a competitive process first, which sits oddly beside the praise that the fleet-orchestration console gave supervisors usable congestion data. Aalen (GRI-01) met integration test in the customer sandbox on 2024-03-24 without a variation to plan. Solveig Aaltonen gave 51 minutes on 2026-04-24 and made one quantified claim, on reduction in temporary labour headcount at peak, of 6 per cent. Aalen logged final acceptance certificate on 2025-02-27; the customer attributes the slip to customer middleware team could not release the interface test window. Solveig Aaltonen would re-buy on the basis of yes, subject to a commercial reset on overage, having first complained that battery swap procedure is awkward for the night crew. Aalen (GRI-01) met integration test in the customer sandbox on 2024-03-24 without a variation to plan. Solveig Aaltonen would re-buy on the basis of yes, subject to a commercial reset on overage, having first complained that battery swap procedure is awkward for the night crew.

Schedule 11.4 - Grindelwald Precision: acceptance certificates and amounts withheld

Site	Conditional acceptance date	Final acceptance date	Punch items at conditional	Punch items open	Withheld amount
Aalen (GRI-01)	2024-10-13	2025-02-27	8	0	nil

Grindelwald Precision - evidence and reliance register

The Grindelwald Precision bundle sets out that the executed protocol and the certificates carry most of the analytical weight, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. Annegret Holzknecht's team goes further: two sources were not relied upon because provenance could not be established. The Grindelwald Precision estate carries the detail at record level deliberately, because an account-level average at Grindelwald Precision conceals the site-to-site variation that decides how this relationship behaves at renewal. Kwesi Boateng-Marsh, the site general manager, spoke on 2026-02-03 and praised that firmware rollbacks were quick when a release misbehaved, answering the re-buy question with yes, subject to a commercial reset on overage. The conditional acceptance certificate in the Grindelwald Precision bundle is cited throughout this review as the evidence for which items were carried forward as punch. Berenice Straatman gave 33 minutes on 2026-02-22 and made one quantified claim, on units picked per operator hour, of 38 per cent. Grindelwald Precision relies on the support ticket export for its own account of the position, and it shows how each incident was classified for severity. Asked what worked, Kwesi Boateng-Marsh said that firmware rollbacks were quick when a release misbehaved; asked what did not, the answer was that on-site engineering days were consumed by defects rather than tuning. Reading the executed acceptance test protocol for Grindelwald Precision is what establishes the tested criteria and the signatures on them. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. The published reference case study for Grindelwald Precision was produced on request and shows what was claimed to the market.

The Aalen operation carries two qualifications: reliance is stated so a reader can see which conclusion moves if a document fails, and dates are document dates and not the dates on which they were provided. Grindelwald Precision also observed, when the schedule was put to the customer, that the strongest documents in this file came from the customer, not from the company, which is consistent with what the documents show without being proved by them. Grindelwald Precision's file accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Grindelwald Precision was asked for the aged receivable listing precisely because it bears on how promptly this customer actually pays. Whatever the company's materials claim, Berenice Straatman described the position on units picked per operator hour as substantiated: figures tie to the customer's own weekly operations pack. The amended order form at Grindelwald Precision carries more weight than any summary, since it shows nothing, because it was never produced. Talking to Bartholomew Nkomo for 53 minutes by video call produced the claim that cost per outbound line improved by 7 per cent, partly substantiated: one of the two sites supports the claim. Grindelwald Precision was asked for the aged receivable listing precisely because it bears on how promptly this customer actually pays. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. The aged receivable listing at Grindelwald Precision carries more weight than any summary, since it shows how promptly this customer actually pays. Whatever the company's materials claim, Berenice Straatman described the position on units picked per operator hour as substantiated: figures tie to the customer's own weekly operations pack.

Schedule 11.5 - evidence relied upon in the Grindelwald Precision account review

Document or source	Provided by	Date	What it shows	Reliance
support ticket export by severity	company support	2026-04-19	how incidents were classified	corroborated against a second source
customer weekly operations pack	customer operations	2026-02-15	the customer's own performance view	relied upon without further work
company reference case study	company marketing	2026-03-17	what was claimed externally	relied upon after recalculation

Renewal mechanics and probability at Grindelwald Precision

Renewal at Grindelwald Precision turns on 2027-08-08 under a structure described in the order form as initial term still running, first renewal not yet in window, so the operative deadline is the notice date of 2027-05-10 rather than the renewal date itself. Probability is recorded in the band 55 to 75 per cent, and the movement expected alongside renewal is expansion order for one additional site under evaluation. A single trigger drives the risk: labour agreement restricts further automation at the site. Zephyrine Beauty Brands began on 2023-04-04 and its structure is described as initial term running; expansion order under negotiation. Asked what worked, Noor Al-Rashidi said that charge management stopped

being a supervisor problem after tuning; asked what did not, the answer was that the roadmap commitments given during the sales cycle slipped a year. Looking across at Arborlight Home, the renewal structure reads as auto-renew annually unless either party gives sixty days notice. Yuki Tanaka-Priest answered on re-buy with yes, subject to a commercial reset on overage and would not be drawn further on price or on term. Thornquist Parcel is banded at 55 to 75 per cent on renewal, the trigger being sponsor departure removed the internal champion. Aurelie Devereaux-Sim answered on re-buy with no, would run a competitive process first and would not be drawn further on price or on term. Ellery Vandersloot sponsors the Arborlight Home deployment, which covers 8 sites in omnichannel home goods retail. Yuki Tanaka-Priest raised the complaint that reporting cannot be reconciled against the customer's own labour standard, and put the improvement in units picked per operator hour at 33 per cent. Looking across at Meridian Fulfilment, the renewal structure reads as auto-renew in twelve-month increments, ninety-day notice. Asked what worked, Antonina Wrzesniewska said that the fleet-orchestration console gave supervisors usable congestion data; asked what did not, the answer was that support handoffs across time zones lose context between shifts. Vantageport Terminals pays on net sixty, a longer cycle than most of the base and one negotiated at signature.

Annegret Holzknecht would not commit to a renewal position and was explicit that procurement intends to test the market before the notice window closes, which is a normal posture and not by itself a warning sign. Credibility of the threat is the real question, and here it partly is: two competitors can serve this building profile, though neither has a reference at comparable fleet density in precision tooling and metrology equipment. Integration work is the main barrier to switching rather than in the hardware, which is a weaker moat than the company's materials imply. Set beside Northriding Automotive, which runs 5 sites at 246 robots, this account is the better documented of the two. Asked what worked, Antonina Wrzesniewska said that the fleet-orchestration console gave supervisors usable congestion data; asked what did not, the answer was that support handoffs across time zones lose context between shifts. Arborlight Home began on 2021-11-22 and its structure is described as auto-renew annually unless either party gives sixty days notice. Berenice Straatman volunteered that the WMS interface behaved predictably once message mapping settled, and the claim on units picked per operator hour was substantiated: figures tie to the customer's own weekly operations pack. Looking across at Ironbark Building Materials, the renewal structure reads as auto-renew annually; one site non-renewed. Berenice Straatman gave 33 minutes on 2026-02-22 and made one quantified claim, on units picked per operator hour, of 38 per cent. Thornquist Parcel began on 2020-04-06 and its structure is described as renewed twice; currently in a twenty-four-month extension. Bartholomew Nkomo gave 53 minutes on 2026-03-16 and made one quantified claim, on cost per outbound line, of 7 per cent. Set beside Ironbark Building Materials, which runs 3 sites at 81 robots, this account is the better documented of the two. Notes taken on 2026-02-06 record Yuki Tanaka-Priest praising that safety record was clean across two shifts and no lost-time incidents beside the complaint that reporting cannot be reconciled against the customer's own labour standard. Ironbark Building Materials sits in building materials wholesale on hardware purchase plus software subscription, with renewal falling on 2026-10-25 and notice due by 2026-07-27.

Notice discipline on the company side was tested by asking when the renewal conversation actually opens, and the answer given was inside the last quarter before notice falls due. The timing leaves little room to recover a problem, particularly where a punch list or a credit dispute is still open, because the customer then holds two levers at once. Marcus Adeyinka-Blau, Regional Sales Director owns the renewal and has a written plan for it, which several other accounts in this pack do not, so the process weakness here is timing rather than ownership. Notes taken on 2026-04-24 record Solveig Aaltonen praising that implementation engineers were candid about what the pilot would not prove beside the complaint that battery swap procedure is awkward for the night crew. Looking across at Arborlight Home, the renewal structure reads as auto-renew annually unless either party gives sixty days notice. Notes taken on 2026-02-22 record Berenice Straatman praising that the WMS interface behaved predictably once message mapping settled beside the complaint that the pilot success criteria were written loosely and then argued over. Vantageport Terminals renews on 2027-07-19, and Marcus Adeyinka-Blau owns that conversation on the company side. Talking to Bartholomew Nkomo for 53 minutes by video call produced the claim that cost per outbound line improved by 7 per cent, partly substantiated: one of the two sites supports the claim. Tidewater Marine Spares pays on net sixty, a longer cycle than most of the base and one negotiated at signature. Asked what worked, Antonina Wrzesniewska said that the fleet-orchestration console gave supervisors usable congestion data; asked what did not, the answer was that support handoffs across time zones lose context between shifts. Set beside Ossuary Ridge Beverage, which runs 2 sites at 43 robots, this account is the better documented of the two. Asked what worked, Aurelie Devereaux-Sim said that the fleet-orchestration console gave supervisors usable congestion

data; asked what did not, the answer was that battery swap procedure is awkward for the night crew. Meridian Fulfilment deploys 267 robots across 6 sites and is banded at 1.8m to 2.2m. Kwesi Boateng-Marsh was reached by on-site walkthrough and gave a re-buy answer of yes, subject to a commercial reset on overage, which sits oddly beside the praise that firmware rollbacks were quick when a release misbehaved.

Deployment sequence and go-live history at Grindelwald Precision

Aalen (GRI-01) took the first site survey on 2023-11-03 against a plan of 2023-10-07, and the 27-day difference was attributed to seasonal peak freeze: no cutovers permitted during the retail quarter. The remaining gates ran in sequence: radio-frequency readiness, interface build against the host warehouse-management system, sandbox integration testing, then a pilot cell before any protocol was executed. Aalen (GRI-01) reached pilot go-live on 2024-06-28, which is outside the window the order form contemplated. Certificates held at Aalen put final acceptance certificate on 2025-02-27 and explain the 18 days by customer middleware team could not release the interface test window. The pilot charter for Grindelwald Precision was produced on request and shows what the parties agreed the pilot had to prove. Aalen (GRI-01) reached network and radio-frequency readiness review on 2024-01-19, status closed, with 0 punch items open at that gate. The credit note sequence for Grindelwald Precision was produced on request and shows what was conceded and in which period. Once Aalen had dealt with seasonal peak freeze: no cutovers permitted during the retail quarter, the site completed acceptance test protocol execution on 2024-09-10. The credit note sequence obtained from Grindelwald Precision was read in full rather than in extract, since it shows what was conceded and in which period. Certificates held at Aalen put network and radio-frequency readiness review on 2024-01-19 and explain the 48 days by customer sponsor on medical leave, sign-off authority unclear. The console meter extract obtained from Grindelwald Precision was read in full rather than in extract, since it shows the volume the company recorded. On the evidence from Aalen, site survey and slotting scan slipped 27 days and the status now reads closed. The published reference case study in the Grindelwald Precision bundle is cited throughout this review as the evidence for what was claimed to the market. Once Aalen had dealt with seasonal peak freeze: no cutovers permitted during the retail quarter, the site completed acceptance test protocol execution on 2024-09-10.

Interface work is where this deployment differed from the company's published sequence, because Grindelwald Precision runs a host system whose release calendar it does not control and would not move. The mapping exercise outran the build effort, and two of the mapping rounds were driven by customer master data on slot dimensions rather than by anything the company controlled. Attribution of the slip turns on that distinction, and it is examined against the documents later in this file rather than settled here. Once Aalen had dealt with seasonal peak freeze: no cutovers permitted during the retail quarter, the site completed site survey and slotting scan on 2023-11-03. Reading the host interface message log for Grindelwald Precision is what establishes how the interface behaved on receipt confirmations. Aalen (GRI-01) met integration test in the customer sandbox on 2024-03-24 without a variation to plan. The credit note sequence obtained from Grindelwald Precision was read in full rather than in extract, since it shows what was conceded and in which period. Aalen (GRI-01) reached punch-list closure review on 2025-01-08, status punch list closed, with 7 punch items open at that gate. Reading the support ticket export for Grindelwald Precision is what establishes how each incident was classified for severity. Planned and actual coincide at Aalen for integration test in the customer sandbox, both falling on 2024-03-24. Grindelwald Precision let us check the site labour standard against the customer's own copy, and it shows the baseline against which benefits are claimed. Acceptance slipped 48 days at Aalen before network and radio-frequency readiness review was recorded as closed. The aged receivable listing for Grindelwald Precision was produced on request and shows how promptly this customer actually pays. Aalen lost time because charger electrical permit issued late by the local authority, and conditional acceptance certificate moved to 2024-10-13 from a planned 2024-09-27.

Matthias Ehrenwerth described the deployment as slower than promised but not badly run, and offered the charger electrical works as an example of a delay the site itself caused and did not blame the vendor for. The vendor record is less generous, recording the same period as awaiting customer readiness without naming the electrical permit, so the two records agree on the fact and disagree on the framing. Both readings are carried forward because the pattern recurs across several accounts in this pack and is itself a finding. On the evidence from Aalen, site survey and slotting scan slipped 27 days and the status now reads closed. Grindelwald Precision relies on the host interface message log for its own account of the position, and it shows how the interface behaved on receipt confirmations. Acceptance slipped 72 days at Aalen

before pilot cell go-live was recorded as closed. The console meter extract at Grindelwald Precision carries more weight than any summary, since it shows the volume the company recorded. Deployment records at Aalen leave site survey and slotting scan closing on 2023-11-03, status closed. The published reference case study in the Grindelwald Precision bundle is cited throughout this review as the evidence for what was claimed to the market. Aalen (GRI-01) reached conditional acceptance certificate on 2024-10-13, status conditional acceptance issued, with 8 punch items open at that gate. The published reference case study for Grindelwald Precision was produced on request and shows what was claimed to the market. Progress at Aalen shows site survey and slotting scan signed off on 2023-11-03 against 2023-10-07 as planned. Grindelwald Precision was asked for the site labour standard precisely because it bears on the baseline against which benefits are claimed. Aalen (GRI-01) cleared punch-list closure review on 2025-01-08 against a plan of 2024-11-06, a difference of 63 days recorded against supplier lead time on the conveyor tie-in guard package.

Site and fleet inventory: Grindelwald Precision

The Grindelwald Precision workpaper evidences that charger bay provision is the constraint at two of the sites, not fleet size, and the schedule below is the evidence for it, drawn from 1 records rather than from a summary prepared for us. Annegret Holzknecht carries a further observation: site codes here follow the company's own numbering so the schedule can be tied back. Diligence on Grindelwald Precision settled on this level of detail for a reason, because an account-level average at Grindelwald Precision conceals the site-to-site variation that decides how this relationship behaves at renewal. Acceptance slipped 48 days at Aalen before network and radio-frequency readiness review was recorded as closed. Attainment figures from Aalen show 96 per cent on response against 88 per cent on resolution. Progress at Aalen shows site survey and slotting scan signed off on 2023-11-03 against 2023-10-07 as planned. Aalen (GRI-01) shows pick confirmation duplicated in the host WMS as the complaint theme, on 28 tickets in 2025-Q3. Turning to Aalen, the tracker puts wms interface build and message mapping at 2024-03-09 where the plan said 2024-02-19, the gap being host WMS upgrade re-sequenced the interface build. Aalen (GRI-01) shows charger handshake failures on one dock as the complaint theme, on 25 tickets in 2024-Q3. The slip of 60 days at Aalen on acceptance test protocol execution is the one the customer raises first, and it traces to seasonal peak freeze: no cutovers permitted during the retail quarter. Although the console reported no outage at Aalen, the site still raised 24 tickets in 2024-Q2.

Matthias Ehrenwerth comes with two limits on its use: coverage descriptions were confirmed verbally and not from a rota document, and robot counts are as at the fieldwork date and move with seasonal flex licences. The Aalen team also observed, when the schedule was put to the customer, that shift coverage rather than robot count drives the support burden at these buildings, which is consistent with what the documents show without being proved by them. The Grindelwald Precision estate accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Punch items numbering 0 remained at Aalen when final acceptance certificate was signed on 2025-02-27. Severity being classified by the vendor, Aalen recorded only 1 events at the top severity in 2025-Q1. Certificates held at Aalen put network and radio-frequency readiness review on 2024-01-19 and explain the 48 days by customer sponsor on medical leave, sign-off authority unclear. Recurring at Aalen through that quarter was wireless roaming loss in the cold zone, across 23 tickets. Turning to Aalen, the tracker puts final acceptance certificate at 2025-02-27 where the plan said 2025-02-09, the gap being customer middleware team could not release the interface test window. Aalen (GRI-01) shows charger handshake failures on one dock as the complaint theme, on 25 tickets in 2024-Q3. Aalen (GRI-01) cleared final acceptance certificate on 2025-02-27 against a plan of 2025-02-09, a difference of 18 days recorded against customer middleware team could not release the interface test window. Aalen (GRI-01) shows pick confirmation duplicated in the host WMS as the complaint theme, on 8 tickets in 2026-Q1.

Schedule 11.6 - Grindelwald Precision: site fleet inventory and shift coverage

Site code	City	Robots	Shift coverage	Charger bays	Go-live
GRI-01	Aalen	23	three shifts, six days	6	2024-01-08